

实验报告

实验名称: 声光效应.

一. 实验数据.

$\delta = 56.0 \text{ mm}$

示波器Y轴灵敏度: 1 V/div

CCD中心位置/mm	声光器件位置/mm	一个周期总小格数/div	未衍射光强/div
100.00	600.00	41	38

$D = 600 - 100 - 56 = 444.0 \text{ mm}$

$P = 100 \text{ mW}$

示波器Y轴灵敏度: 1 V/div

测量次数	1	2	3	4	5	6	7
超声波频率 f_s / MHz	80.05	85.03	90.09	95.01	100.08	105.08	110.04
偏转格数 d_0 / div	7.5	8.0	8.4	9.0	9.4	9.9	10.2
	8	9					
f_s / MHz	115.06	120.03					
d_0 / div	10.5	11.0					

$P = 150 \text{ mW}$

测量次数	1	2	3	4	5	6	7	8	9
超声波频率 f_s / MHz	80.0	90.0	100.0	110.0	120.0	130.0	140.0	150.0	160.0
衍射光强 I_1 / div	25.1	27.3	35.0	35.7	26.0	28.5	36.7	35.1	37.1

$$f = 107.5 \text{ MHz}$$

超声波功率 P/mW	50	150	250	350	450	550	650	750	850	950
透射光强 I_0/div	38.0	35.3	28.8	15.1	14.8	13.0	12.0	11.0	9.8	7.5
衍射光强 I_1/div	0	2.9	11.6	24.0	28.5	32.1	34.2	35.1	36.5	37.2

二. 数据处理

1. 定标.

CCD 波形 - 帧对应 41 个小格.

则每小格对应的实际空间距离为 $\Delta s = \frac{1}{41} \times 2700 \times 11 \text{ mm} = 0.7244 \text{ mm}$.

2. 布拉格衍射的声光偏转.

根据 $l = d_0 \cdot \Delta s$ 空气中偏转角度 $\varphi \approx \frac{l}{D}$.

d_0/div	7.5	8.0	8.4	9.0	9.4	9.9	10.2	10.5	11.0
l/mm	5.433	5.792	6.085	6.516	6.804	7.171	7.388	7.602	7.968
$\varphi / \times 10^{-3} \text{ rad}$	12.24	13.05	13.70	14.68	15.34	16.15	16.64	17.13	17.95

使用一元线性回归法:

$$\text{设 } y = \hat{\beta}x + \hat{\alpha} \quad y = f(x) \quad x = \varphi.$$

$$\text{则 } \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i = 15.21 \times 10^{-3} \text{ rad} \quad \bar{y} = \frac{1}{n} \sum_{i=1}^n y_i = 100.05 \text{ MHz}.$$

$$\overline{x^2} = \frac{1}{n} \sum_{i=1}^n x_i^2 = 234.6644 \times 10^{-6} (\text{rad}^2) \quad \overline{y^2} = \frac{1}{n} \sum_{i=1}^n y_i^2 = 10177.09 (\text{MHz})^2$$

$$\overline{xy} = \frac{1}{n} \sum_{i=1}^n x_i \cdot y_i = 1545.25 \times 10^{-3}$$

$$\text{则 } \hat{\beta} = \frac{\bar{x} \cdot \bar{y} - \overline{xy}}{\overline{x^2} - \bar{x}^2} = \frac{15.21 \times 10^{-3} \times 100.05 - 1545.25 \times 10^{-3}}{(15.21 \times 10^{-3})^2 - 234.6644 \times 10^{-6}} = 7.0745 \times 10^3$$

$$\hat{\alpha} = \bar{y} - \hat{\beta} \bar{x} = 100.05 - 7.0745 \times 10^3 \times 15.21 \times 10^{-3} = -7.5531$$

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则线性回归方程为 $y = 7.0745 \times 10^3 x - 7.5531 \text{ MHz}$.

相关系数 $r = \frac{\overline{xy} - \bar{x} \cdot \bar{y}}{\sqrt{(\bar{x}^2 - \bar{x}^2)(\bar{y}^2 - \bar{y}^2)}} = 0.99727$ f_s 与 φ 有较强的线性关系.

设衍射角为 ϕ . 根据 $\phi = 2\pi \frac{\lambda}{\lambda_s} \approx \frac{\lambda_0}{n v_s} \cdot f_s$.

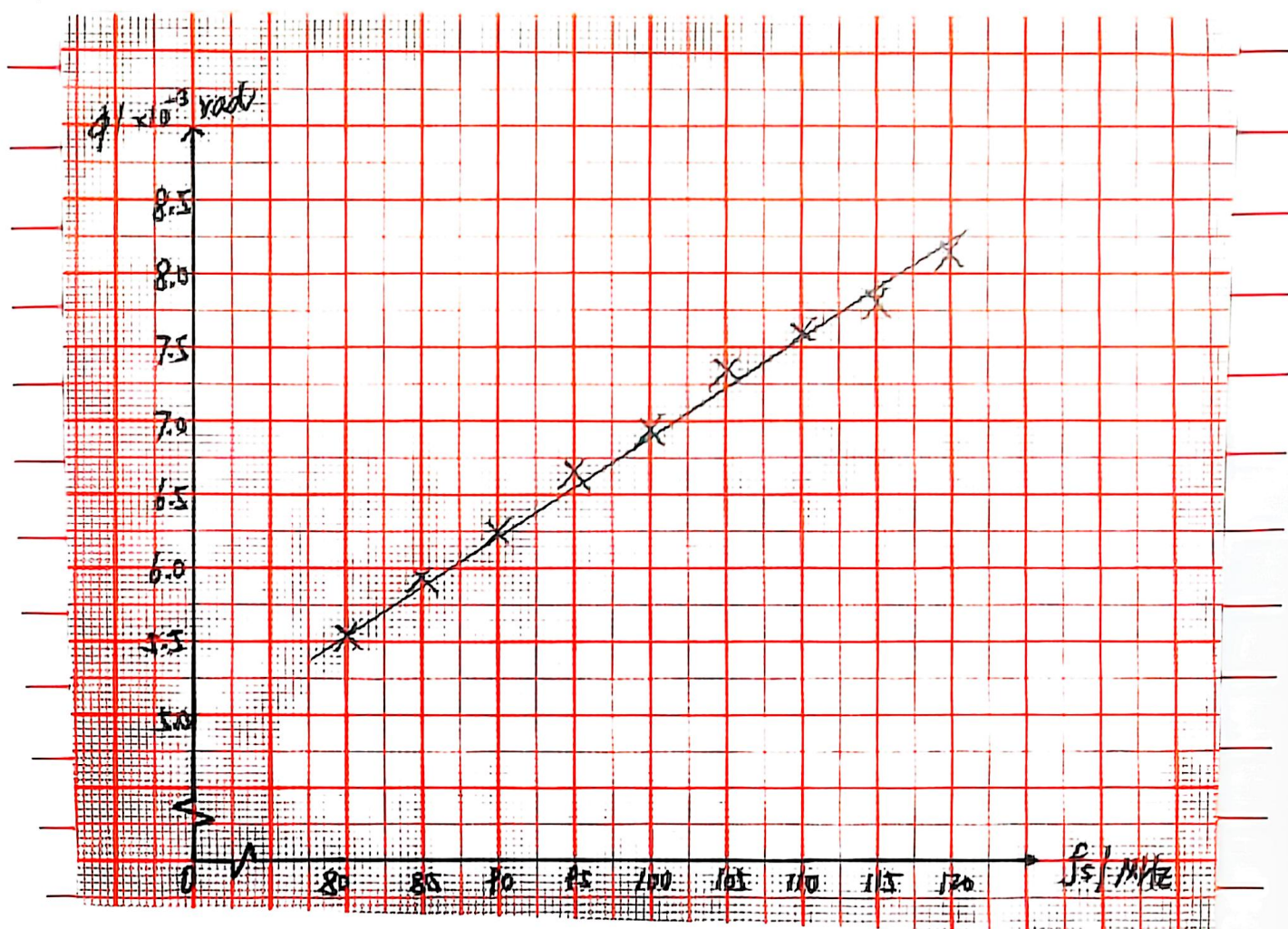
再由 $\frac{\partial \phi}{\partial \sin \phi} = \frac{\phi}{\sin \phi} = \frac{1}{n} \quad n \text{ 取 } 2.2$.

$\therefore v_s = \frac{\lambda_0}{n \phi} \cdot f_s = \frac{\lambda_0}{\phi} f_s = 650 \times 10^{-9} \times 7.0745 \times 10^3 \times 10^6 = 4598.425$.

相对误差 $y = \left| \frac{4598.425 - 4200}{4200} \right| \times 100\% = 9.49\%$.

$\varphi / \times 10^{-3} \text{ rad}$	12.24	13.05	13.70	14.68	15.34	16.15	16.64	17.13	17.75
$\phi / \times 10^{-3} \text{ rad}$	5.56	5.93	6.23	6.67	6.97	7.34	7.56	7.79	8.16

ϕ 与 f_s 曲线如下图所示.



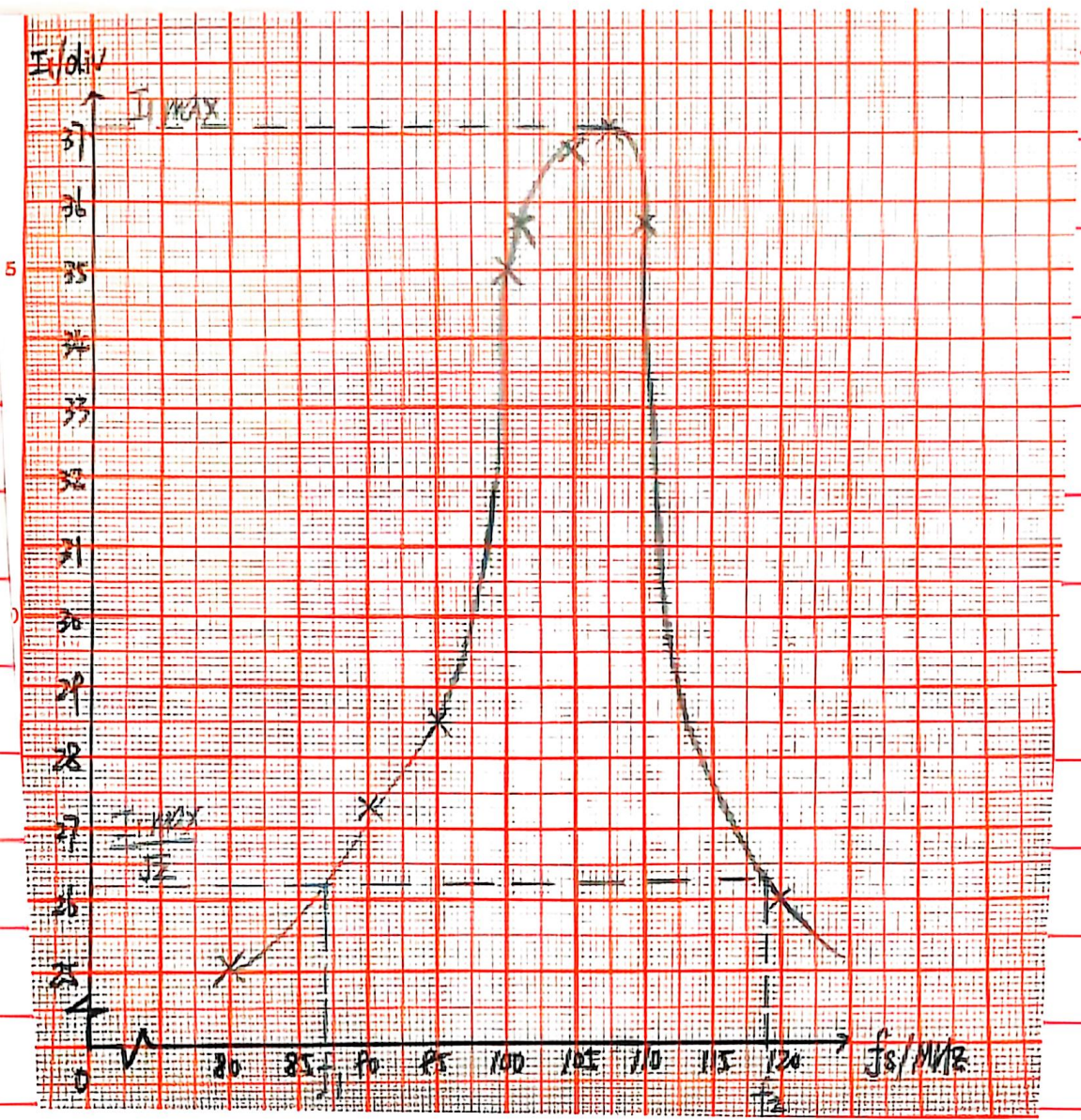
3. 测量中心频率与带宽.

由数据可知 当 $f_s = 107.5 \text{ MHz}$ 时 I 最大为 37.1 格 即中心频率为 $f_0 = 107.5 \text{ MHz}$

f_s 与 I 的关系曲线如下图所示

当偏转为 $\frac{37.1}{\sqrt{2}} = 26.23$ 格时, 对应频率大致为 $f_1 = 87.0 \text{ MHz}$ $f_2 = 119.1 \text{ MHz}$

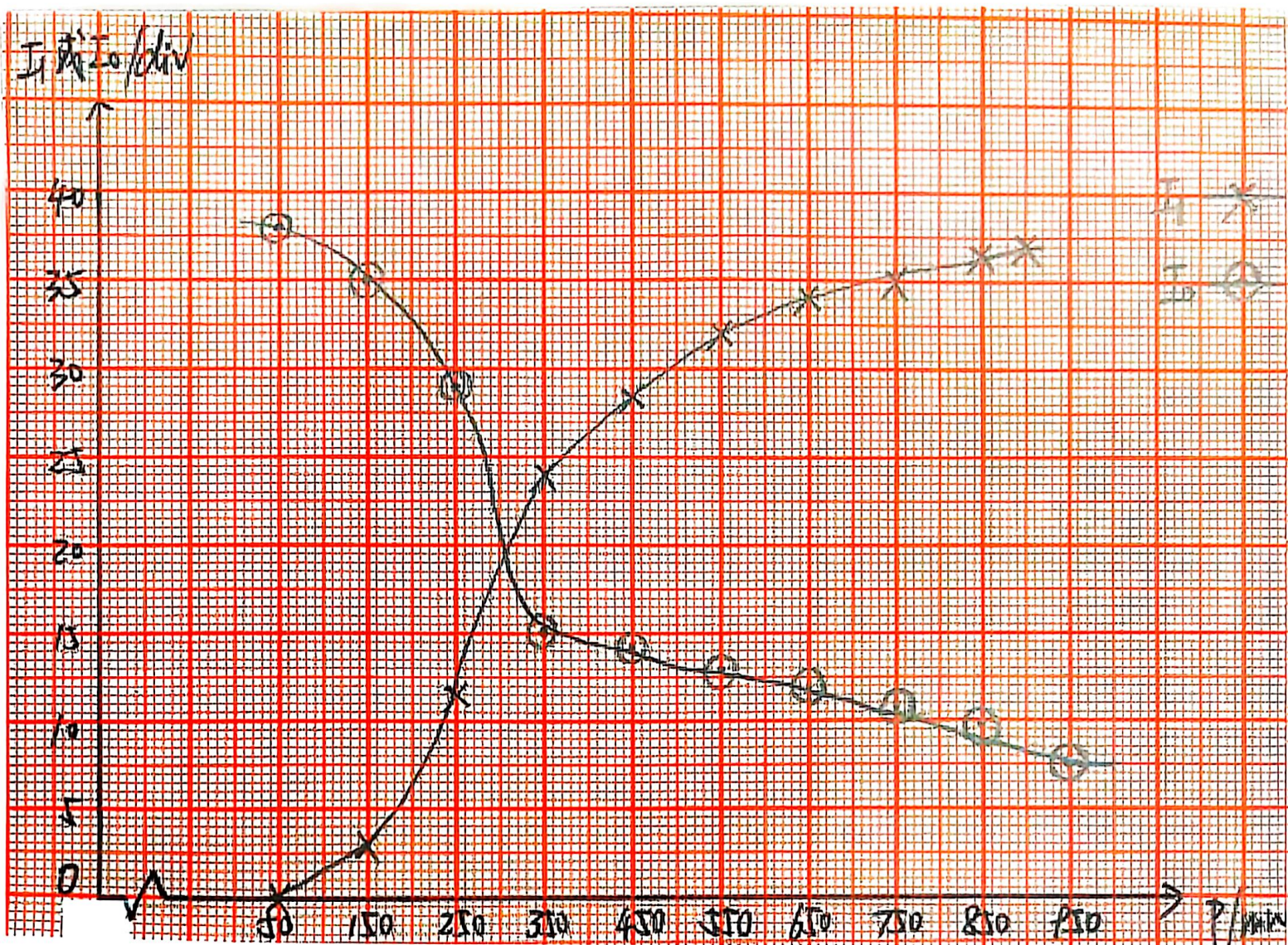
$\therefore$ 带宽 $\Delta f_s = f_2 - f_1 = 32.1 \text{ MHz}$.



4. 声光调制. ($f_c = 107.5 \text{ MHz}$).

根据数据绘出声光调制曲线. 如下图所示.

最大衍射效率. $\eta_{\max} = \frac{I_{1\max}}{I_0} \times 100\% = \frac{37.2}{38.0} \times 100\% = 97.89\%$.



5. 拉曼-奈斯衍射

$$f_c = 107.5 \text{ MHz} \quad v_s = 4200 \text{ m/s} \quad n = 2.2 \quad I_0 = 37.0 \text{ div} \quad I_1 = 10.1 \text{ div} \quad \Delta \text{小格} = 23 \text{ div}$$

$$\text{理论值} \quad \sin \phi_0 = \frac{m \lambda_0}{n \lambda_s} = \frac{\lambda_0 f_c}{n v_s} = \frac{650 \times 10^{-9} \times 107.5 \times 10^6}{2.2 \times 4200}$$

$$\Rightarrow \phi_0 = 0.007562 \text{ rad}$$

$$\text{偏转角} \phi. \quad \tan \phi = \frac{t}{2D} = \frac{0.7244 \times 23}{2 \times 440}$$

$$\Rightarrow \phi = 0.04166 \text{ rad}$$

$$\text{衍射角} \quad \frac{\sin \phi}{\sin \phi_0} = \frac{n}{1} = \frac{2.2}{1}$$

$$\Rightarrow \phi = 0.008605 \text{ rad} \quad \text{相对误差} \gamma = \left| \frac{\phi - \phi_0}{\phi_0} \right| = 13.71\%$$

$$\text{最大衍射效率} \quad \eta = \frac{I_1}{I_0} \times 100\% = \frac{10.1}{37.0} \times 100\% = 27.30\% < 47.81\%$$

6. 思考题

1. 当超声波在介质中传播时, 会引起介质的弹性应变在时间和空间上做周期性变化, 从而导致介质的折射率也发生相应的变化. 当光束通过有超声波的介质后就会产生衍射现象. 有超声波传播着的介质存在附加的相位延迟, 周期性变化, 如相位光栅一般.
2. 先粗调平台, 再微调介质方向, 从而改变激光入射角度, 直至衍射图样对称, 即 0 级衍射光光强基本一致.
3. 可能应用: 制作声光调制器件, 声光偏转器件, 可调谐滤光器, 光开关等.

$$\frac{546 \times 10^3 \times 5 \times 10^6}{0.7 \times 2700 \times 11} = 5.443 \times 10^9 \quad k = 5.443 \times 10^9$$

$$v = \frac{\lambda_0}{\lambda_s} = \frac{\lambda_0 \cdot f}{\phi} = \lambda_0 \cdot k = 3537.7$$

CCD 位置 /mm 100.00

$$y = \frac{\phi_{200} - 3537.7}{\phi_{200}} \times 100\% = 15.8\%$$

声光器件位置 /mm 600.00

$$l_1 = \frac{8.0}{41.6} \times 2700 \times 11$$

点分辨率 /div 41.0

$$\phi_1 = \frac{l_1}{L}$$

未衍射光强 I_0 /div 38

$$\phi_2 = \frac{l_2}{L}$$

$$L = 556 \text{ mm}$$

$$D = 600 - 100 = 500 \quad 600 - 100 - 56 = 444$$

$$k = \frac{\phi_2 - \phi_1}{f_2 - f_1} = \frac{l_2 - l_1}{L(f_2 - f_1)}$$

$$L = 444 \text{ mm}$$

$$= \frac{1}{41.6 \times 2700 \times 11}$$

$$P = 100 \text{ mW}$$

y轴: 1V/div

$$556 \times 10^3 \times 10.$$

$$120.0271$$

次数 1 2 3 4 5 6 7 8 11.0

f_s /MHz 80.0503 85.0254 90.0945 95.0106 100.0761 105.0771 110.0403 115.0582

d_0 /div 7.5 8.0 8.4 9.0 9.4 9.9 10.2 10.5

$$P = 150 \text{ mW}$$

$$107.5$$

次数

$$37.1$$

f_s /MHz 80.0 90.0 100.0 110.0 120.0 130.0 140.0 150.0 160.0

I_1 /div 25.1 27.3 35.0 35.7 26.0 28.5 35.3 35.1

$$P = 150 \sim 1000 \text{ mW} \quad f = 107 \text{ MHz}$$

P /mW 50 150 250 350 450 550 650 750 850 950

I_0 /div 38.0 35.3 28.8 15.1 14.8 13.0 12.0 11.0 9.8 7.5

I_1 /Div 0 2.9 11.6 24.0 28.5 32.1 34.2 35.1 36.5 37.2

$$\frac{466 \times 10^3 \times 5 \times 10^6}{0.5 \times 2700 \times 11}$$

$$= 6.129 \times 10^9$$

$$v = \lambda_0 \cdot k = 3983.85$$